

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (9 pts) Express each sentence in sentential logic using proper notation:

a) Either A or both B and C.

$$A \vee (B \wedge C)$$

b) A exactly when B.

$$A \leftrightarrow B$$

c) Exactly one of A or B.

$$\sim (A \leftrightarrow B) \quad \text{or} \quad (A \wedge \sim B) \vee (\sim A \wedge B) \quad \text{or} \quad (A \vee B) \wedge [\sim (A \wedge B)]$$

$$\text{or} \quad (A \vee B) \wedge [\sim (A \wedge B)]$$

2. (6 pts) Consider the five basic truth functions $f_{\sim}$, $f_{\wedge}$, $f_{\vee}$, $f_{\rightarrow}$, and $f_{\leftrightarrow}$, to compute the value of each composite function below. Show some steps. Do not just write down a value.

a) $f_{\sim}(f_{\vee}(T, F), f_{\wedge}(T, F)) = f_{\rightarrow}(T, F) = F$

b) $f_{\sim}(f_{\leftrightarrow}(T, f_{\vee}(F, T)) = f_{\sim}(f_{\leftrightarrow}(T, T)) = f_{\sim}(T) = F$

3. (5 pts) Prove the argument $p \vee q, p \therefore q$ is invalid.

p	q	$p \vee q$	$(p \vee q) \wedge p$	$[(p \vee q) \wedge p] \rightarrow q$
T	T	T	T	T
T	F	T	T	F
F	T	T	F	T
F	F	F	F	T

In the second line both premises are true but the conclusion is false so this is not a valid rule of deduction

4. (18 pts) Answer the following questions about standard truth tables.

a) How many lines will there be in the standard truth table with 4 variables?

$$2^4 = 16$$

b) What is a sentence called if it has truth value T for every line of the table?

Tautology

c) What is a sentence called if it has truth value F for every line of the table?

Contradiction

d) What is a sentence called if it has truth value T for some lines and F for other lines?

Contingency

e) What does it mean if two sentences, A and B, are logically equivalent?

The columns for the two sentences are identical

f) What is the maximum number of statements with 3 variables such that no two of the statements are logically equivalent?

Each column has 8 entries. Each entry can be T or F

so 2^8 different columns are possible.

$$2^8 = 256$$

5. (4 pts) Find a formal sentence using only the connectives $\sim$ and $\wedge$ that is logically equivalent to the sentence $\sim(p \rightarrow q)$

$$\sim(p \rightarrow q) \equiv \sim((\sim p) \vee q) \equiv (\sim(\sim p) \wedge (\sim q)) \equiv p \wedge (\sim q)$$

$$p \rightarrow q \equiv (\sim p) \vee q \quad \text{so}$$

T
F
T
T

6. (4 pts) Construct a formal sentence B that satisfies the following truth table.

p	q	r	B
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

$$B = (p \wedge q \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r)) \vee ((\neg p) \wedge (\neg q) \wedge (\neg r))$$

simplifies to

$$(p \wedge q \wedge r) \vee ((\neg p) \wedge (\neg r))$$

7. (8 pts) Rewrite each statement and its contrapositive as an if-then statement.

a) Centre wins when they score at least 70 points.

Rewrite: If Centre scores at least 70 points, then they win.

Contrapositive:

If Centre doesn't win, then they score less than 70 points.

b) Every differentiable function is continuous.

Rewrite: If f is differentiable, then f is continuous.

Contrapositive:

If f is not continuous, then f is not differentiable.

8. (8 pts) Construct a five-step proof to show $A \rightarrow (\sim C), B \wedge C \therefore \sim A$ is a valid argument. Label each step using the proper name for the rules of deduction used.

1. $B \wedge C$ premise
2. C follows from 1 by conjunctive simplification
3. $A \rightarrow (\sim C)$ Premise
4. $\sim(\sim C)$ follows from 2 by double negation
5. $\sim A$ follows from 3 and 4 by modus tollens

9. (10 pts) Prove the statement: If $a \equiv b \pmod{n}$ and x is any integer, then $ax \equiv bx \pmod{n}$.

For any integers a, b, n , and x with $n > 0$,

Let $a \equiv b \pmod{n}$.

By the test for congruence this means $a - b = nk$ for some $k \in \mathbb{Z}$.

Multiply through by x to get $(a - b)x = nkx$.

So $ax - bx = n \cdot (kx)$ is a multiple of n since $kx \in \mathbb{Z}$.

Therefore by the test for congruence we know

$$ax \equiv bx \pmod{n}.$$

□

10. (6 pts) Consider the set $\{a, b, c, d, e, f, g, h, i, j\}$ with subsets $A = \{a, e, i\}$, $B = \{b, c, d, e, g\}$, and $C = \{b, d, f, j\}$. List the elements in each of the following sets. Use proper notation.

a) $A \times A = \{(a, a), (a, e), (a, i), (e, a), (e, e), (e, i), (i, a), (i, e), (i, i)\}$

b) $C \setminus B = \{f, j\}$

c) $B^c \cup A = \{a, f, h, i, j\} \cup \{a, e, i\} = \{a, e, f, h, i, j\}$

11. (10 pts) Prove or disprove the statement: If x^2 is an irrational number then x is also an irrational number.

By contrapositive we show if x is rational, then x^2 is rational.

Let x be in $\mathbb{Q}$. This means $x = \frac{a}{b}$ where a, b are integers and $b \neq 0$

Therefore $x^2 = \frac{a^2}{b^2}$ is rational since a^2 and b^2 are integers and b^2 is not 0 by the zero product property.



12. (8 pts) Write both of the theorems below as **if-then sentences**.

a) The zero product property:

For any two real numbers x and y

If $x \cdot y = 0$ then $x = 0$ or $y = 0$.

b) The division algorithm:

If a and n are integers with $n > 0$

then there exist unique integers q and r with

$0 \leq r < n$ such that $a = nq + r$

13. (4 pts) List all the numbers between -10 and 20 that are congruent to 5 mod 7

$\{-9, -2, 5, 12, 19\}$